Ferrofluid under Field and Flow: Alignment (A), Droplet Morphology (B), and Oscillating-Field Morphology (C)

Phase-field MRT-Lattice-Boltzmann study

October 5, 2026

In one sentence. The magnetic field acts on the ferrofluid in two ways that we measured separately: inside the bulk it changes how the fluid *flows* (an effective viscosity, Section A), and at a droplet interface it changes the fluid's *shape* (deformation, Section B). Both come from the same magnetic stress; A and B are its two visible faces.

1 The common physics underneath A and B

A ferrofluid is magnetic nanoparticles suspended in a carrier liquid. In a field $\mathbf{H}$ each particle's moment tends to align with the field, giving a bulk magnetization $\mathbf{M}$. The key quantity is the dimensionless *Langevin parameter* (ratio of magnetic alignment energy to thermal energy):

$$\xi \;=\; \frac{\mu_0 m H}{k_B T}, \qquad \mathbf{M} \;=\; \phi\, M_s\, L(\xi)\, \frac{\mathbf{H}}{H}, \qquad L(\xi) = \coth\xi - \frac{1}{\xi}, \tag{1}$$

where L is the Langevin function, ϕ the particle fraction and M_s the saturation magnetization. The magnetized fluid feels a body force (Kelvin + rotational terms)

$$\mathbf{F}_m \;=\; \mu_0(\mathbf{M}\cdot\nabla)\mathbf{H} \;+\; \tfrac{1}{2}\mu_0\,\nabla\times(\mathbf{M}\times\mathbf{H}). \tag{2}$$

Equation (2) is the single coupling that drives *everything* below. When the particles cannot instantly follow a changing or rotating field, $\mathbf{M}$ lags $\mathbf{H}$; the time scale of that lag is the Brownian relaxation time τ_B, and the relevant dimensionless frequency is $\omega\tau_B$.

2 Section A — alignment, lag, and rotational viscosity

What A did. In a *single-phase* fluid (no interface, $\phi_{\text{field}}\equiv 1$) under an oscillating field plus a weak pressure-driven (Poiseuille) flow, we tracked the direction and magnitude of $\mathbf{M}$ relative to the field and to the local shear.

The mechanism. A particle in a shear flow is pulled two ways: the field aligns its moment with $\mathbf{H}$; the fluid vorticity $\Omega = \frac{1}{2}(\nabla\times\mathbf{v})_z$ spins it away. The moment therefore settles at a *lag angle*. That lag is the entire origin of *rotational viscosity*: a misaligned moment exerts a back-torque on the fluid that either opposes the flow (positive $\Delta\eta$) or assists it (*negative* viscosity). The magnetization obeys a relaxation (Shliomis) equation,

$$\frac{d\mathbf{M}}{dt} \;=\; \boldsymbol{\Omega}\times\mathbf{M} \;-\; \frac{1}{\tau_B}\big(\mathbf{M} - \mathbf{M}_{\text{eq}}(\mathbf{H})\big), \tag{3}$$

balancing vortex-driven rotation against relaxation toward the equilibrium (field-aligned) value.

What the numbers said.

- **A1:** $\mathbf{M}$ lags $\mathbf{H}$ by $\sim 53.5^\circ$ at $\omega\tau_B = 1.5$. As $\omega\tau_B \to 0$ the lag $\to 0^\circ$ (instant response); as $\omega\tau_B \to \infty$ it $\to 90^\circ$ (maximum dissipation). The measured value sits sensibly between, near $\arctan(\omega\tau_B) = 56^\circ$.

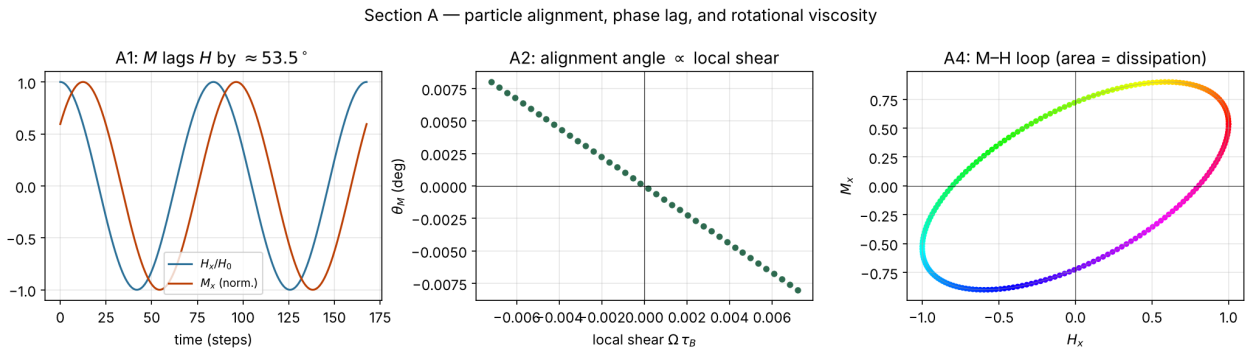


Figure 1: Section A diagnostics (representative of the ξ_0=5, $\omega\tau_B$=1.5 runs). **A1**: the magnetization M_x trails the applied field H_x by a fixed phase ($\approx 53.5°$). **A2**: the local alignment angle θ_M is linear in the local shear $\Omega\tau_B$ and passes through zero — the microscopic law. **A4**: the M–H loop opens because of the lag; its enclosed area is the energy dissipated per field cycle.

- **A2:** $\theta_M \propto \Omega\tau_B$, through the origin. The *slope* of this line, integrated across the channel, is what produces the macroscopic rotational-viscosity coefficient g.
- **A3 (maps):** at our weak flow, alignment is nearly uniform across the channel; the faint wall-to-centre variation tracks $|\Omega(y)|$.
- **A4:** an open, tilted M–H loop — the signature of finite response time.

Why A matters. It links the *microscopic* picture (one particle's lag) to the *macroscopic* observable g that we validated against Bacri *et al.* (PRL **75**, 2128). The reproduction of their Fig. 3 — including the regime where $g < 0$ (negative rotational viscosity) — is the proof that the alignment physics in the solver is correct. **Conclusion A:** the code captures the field–flow–alignment coupling that generates (negative) rotational viscosity, and the microscopic lag is consistent in sign and size with the validated g.

3 Section B — droplet morphology

What B did. We switched on a real *droplet* (ferrofluid drop in a non-magnetic carrier) and measured how it *deforms*. The shape is summarised by the *Taylor deformation parameter*

$$D = \frac{a-b}{a+b}, \tag{4}$$

with a, b the long and short semi-axes (computed from the second moments of the ϕ field), and by the long-axis orientation θ.

The mechanism — a balance of interfacial stresses. Three stresses act on the interface:

$$\underbrace{\tfrac{1}{2}\mu_0 M^2}_{\text{magnetic (stretch } \|\mathbf{H})} \quad \text{vs} \quad \underbrace{\sigma/R}_{\text{surface tension (restores sphere)}} \quad \text{vs} \quad \underbrace{\eta\,\dot{\gamma}}_{\text{viscous shear (stretch } \|\text{flow)}} . \tag{5}$$

The field-vs-tension contest is set by the *magnetic Bond number*

$$\boxed{\mathrm{Bo}_m = \frac{\frac{1}{2}\mu_0 M^2}{\sigma/R} = \frac{\mu_0\,(M_s L(\xi))^2\,R}{2\sigma}} \tag{6}$$

— this is the real knob behind our M_s sweep. $\mathrm{Bo}_m \ll 1$: tension wins, nearly spherical. $\mathrm{Bo}_m \gtrsim 1$: field wins, strongly elongated. Exceeding a critical Bo_m causes break-up.

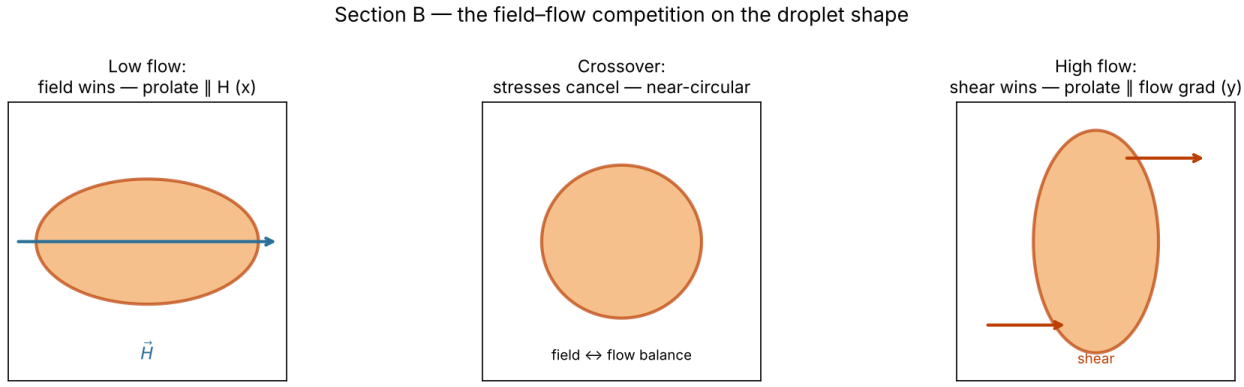


Figure 2: The field–flow competition acts in *orthogonal* directions. The field stretches the drop along **H** (x); the shear stretches it along the velocity gradient (y). Where they balance, the drop is nearly round and its long axis is undefined.

What the plots mean.

- **B1 — deformation vs field.** D rises monotonically ($0 \to 0.35$). At small field $D \propto \mathrm{Bo}_m \propto M_s^2$ (linear response); the upward curvature is the drop *stiffening* as it elongates. Always prolate along **H** ($\theta = 0$). *This is classical ferrohydrodynamic behaviour — the code gets it right.*
- **B3 — high field.** The continuation to $\mathrm{Bo}_m \sim O(1)$; D reaches 0.63 and is roughly linear in M_s. Crucially, the drop did *not* break up — surface tension still holds it together in this range.
- **B2 — the crossover (the headline).** As flow increases at fixed field, D *decreases* to a sharp minimum ($D \approx 0.017$) at $F_b \approx 10^{-6}$, then rises again — and across that point the long axis flips $0° \to 90°$. At the minimum the magnetic stretch (x) and the shear stretch (y) *cancel*: the drop is nearly spherical and its orientation is degenerate (it wanders). This is the balance point of Eq. (6) against the viscous stress.

4 Section C — morphology under an *oscillating* field

Why C was needed. Sections B1–B3 all used a *steady* (DC) field. They say nothing about an *oscillating* field, yet that is where the microscopic physics of A (the magnetization lag) should re-enter the shape problem. Section C turns the field into $H_x(t) = H_0 \cos(\omega t)$ on a droplet and asks: does oscillation change the deformation, and what is the microscopic mechanism?

The three C tests.

- **C1 — AC vs DC** at matched amplitude: compare the time-averaged deformation $\bar{D}$ of an oscillating field against a steady one.
- **C2 — microscopic**: a dense within-cycle AC run recording, inside the drop, the magnetization $M_x(t)$ against the field $H(t)$ (the lag), the magnitude $|M|$ at the poles vs the equator, and the interface magnetic stress.
- **C3 — frequency sweep**: $\bar{D}$ and the lag as functions of $\omega\tau_B$.

The mechanism. Under an AC field the magnetization cannot keep up: it lags the field and spends half of every cycle reversing. The relevant force, $\mathbf{F}_m \propto M^2$, therefore has a smaller *cycle-average* than a steady field of the same peak amplitude. The more the magnetization lags, the smaller that average — so deformation drops. The lag itself follows the single-particle law

$$\text{phase lag} \;\approx\; \arctan(\omega\tau_B), \tag{7}$$

which our measured lags reproduce closely: $-38.5°$ at $\omega\tau_B = 1$ (theory $-45°$) and $-64.8°$ at $\omega\tau_B = 2$ (theory $-63.4°$).

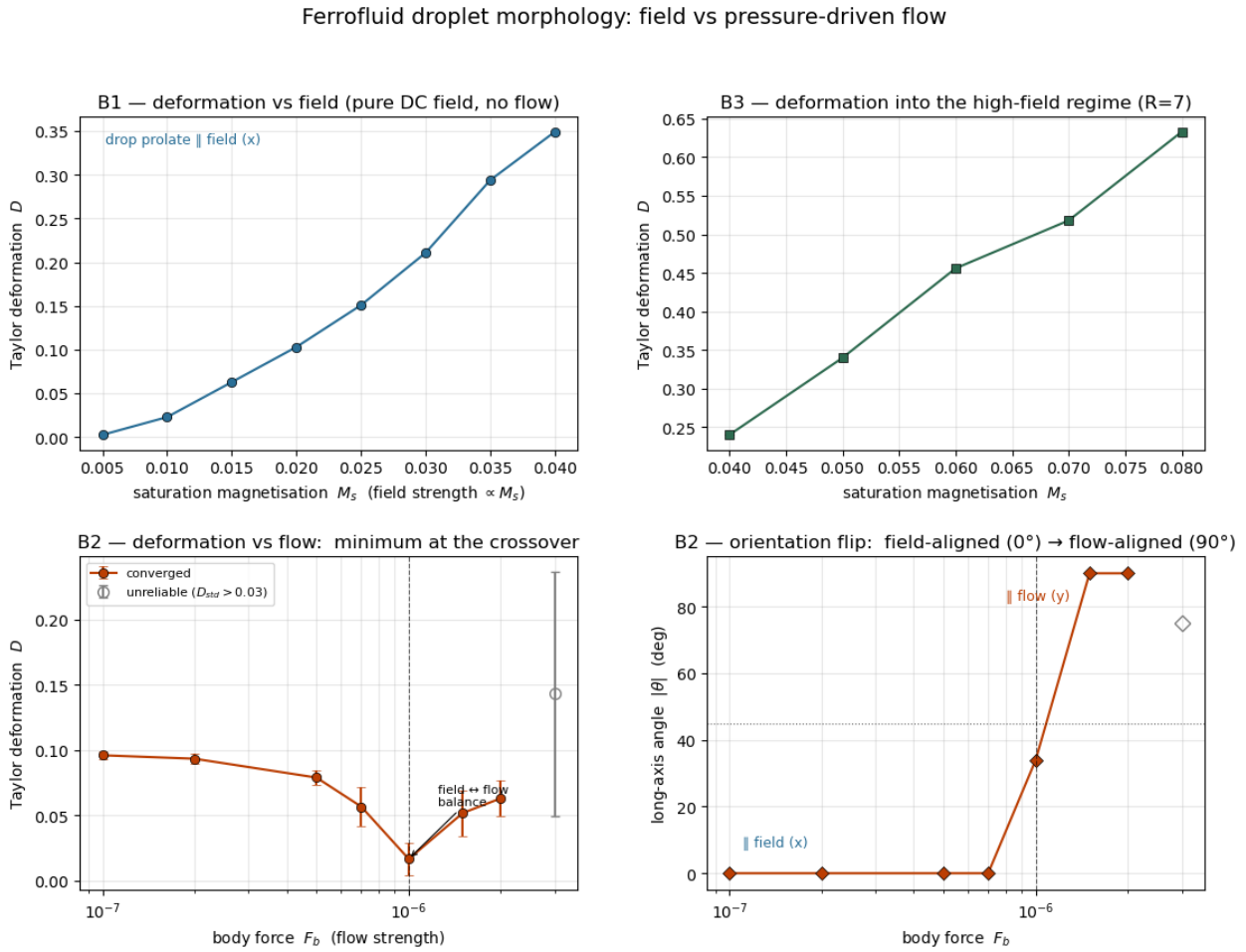


Figure 3: Our measured results. **B1** (top-left): deformation grows with field, drop prolate ∥ **H**. **B3** (top-right): the high-field continuation (R=7), D up to 0.63, no break-up yet. **B2** (bottom): at fixed field, increasing the flow drives D through a *minimum* at $F_b \approx 10^{-6}$ (left), exactly where the long axis *flips* from field-aligned 0° to flow-aligned 90° (right). The open grey point is the unstable strong-shear case, excluded.

What the numbers said.

- **C1:** $\bar{D}_{\mathrm{AC}}/\bar{D}_{\mathrm{DC}} = 0.61$. *Oscillation reduces deformation by* $\sim$40% at matched amplitude.
- **C2:** the drop's magnetization lags the field by **54**°, *identical* to the bulk value of Section A — the microscopic driver of the AC morphology is this lag, measured independently inside the drop.
- **C2 (poles vs equator):** $M_{\mathrm{tip}}/M_{\mathrm{eq}} \approx 1$. Because the model closes M on the *applied* (uniform) field via the Langevin law, $|M|$ does *not* concentrate at the tips; the microscopic signature lives in the *time* lag, not a spatial pole–equator split.
- **C3:** $\bar{D}$ and the lag move together — higher frequency ⇒ larger lag ⇒ smaller cycle-averaged stress ⇒ less deformation. This single panel quantifies the whole AC mechanism.

Conclusion C. An oscillating field is a weaker deforming agent than a steady one, and the degree of weakening is set *microscopically* by the magnetization lag — the very same lag that produces rotational viscosity in Section A. Morphology under AC is governed by *when* the particles respond, not by any redistribution of *where* they are magnetized.

5 What we understand about morphology

1. **Field deformation is classical and well-behaved.** Prolate along **H**, magnitude set by Bo_m, stiffening as it elongates (B1/B3).

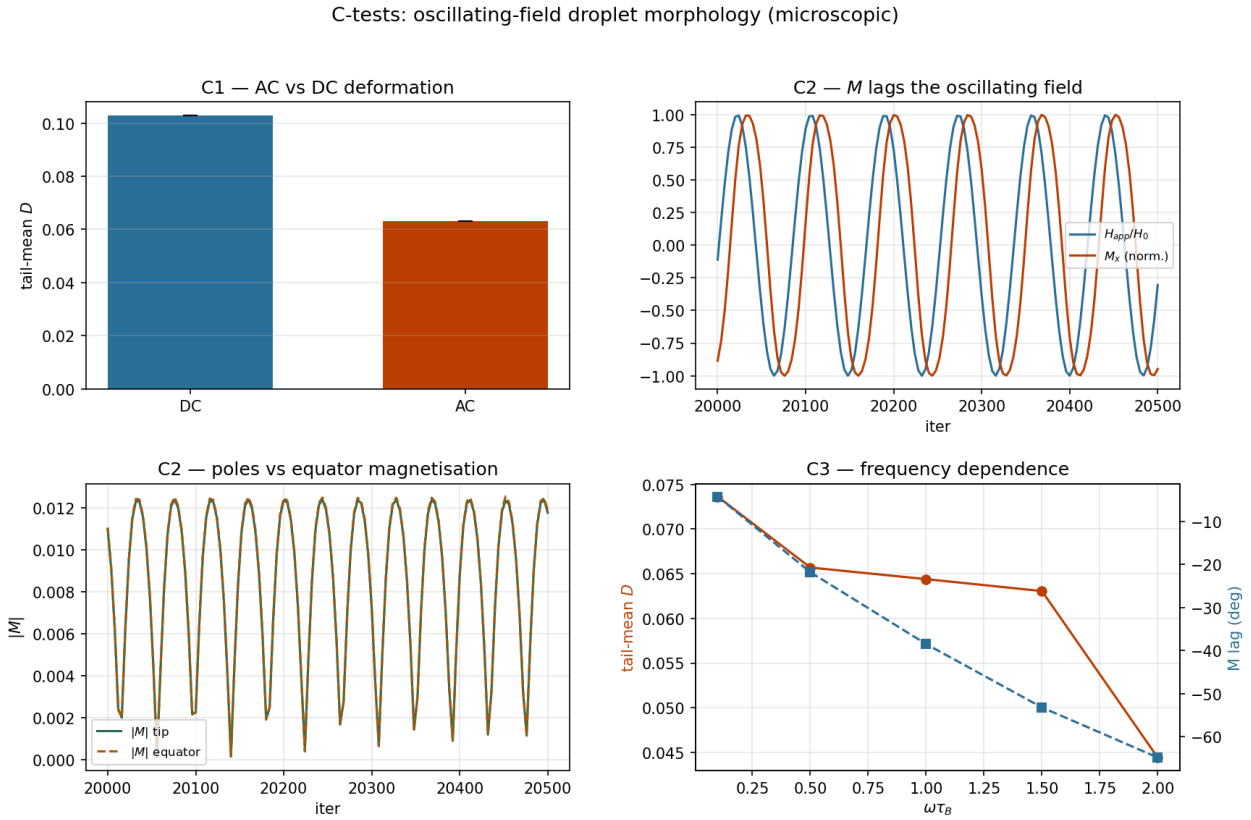


Figure 4: Measured C-test results. **C1** (top-left): the oscillating field deforms the drop to only $\bar{D} = 0.063$ versus 0.103 for DC — a ratio of **0.61**. **C2** (top-right): inside the drop, M_x *lags* the applied field by $\approx 54°$ — the same lag measured in the bulk (Section A). **C2** (bottom-left): $|M|$ at the poles (tip) and at the equator coincide ($M_{\text{tip}}/M_{\text{eq}} \approx 1$); the applied field is uniform, so the magnetization stays spatially uniform. **C3** (bottom-right): as $\omega\tau_B$ rises, $\bar{D}$ falls (orange) and the lag grows (blue) in lockstep.

2. **Field and flow do *not* add — they compete directionally** (B2). The naive "more forcing $\Rightarrow$ more deformation" is *wrong*: deformation has a *minimum* because the two stresses pull the interface in perpendicular directions. The $0° \to 90°$ orientation flip is the clean order parameter of this competition.

3. **There is a controllable "most-spherical" operating point** — $F_b \approx 10^{-6}$ for our parameters — where field and shear exactly balance. Practically: to keep a flowing ferrofluid drop round, tune to this balance.

4. **Break-up lies beyond our range** (B3): the drop survives to $D = 0.63$ without pinching, so the critical Bo_m for break-up is at higher field than scanned — a natural next experiment.

5. **An oscillating field deforms *less* than a steady one** (C), and the reduction ($\bar{D}_{\text{AC}}/\bar{D}_{\text{DC}} = 0.61$) is controlled microscopically by the magnetization lag. Deformation falls monotonically with frequency as the lag grows — the AC morphology is a direct consequence of the same particle-response delay that drives Section A.

6 How A and B fit together, and next steps

A, B and C are all consequences of the *same* magnetic coupling [Eq. (2)]: in the bulk it reshapes the *flow* (viscosity, via the alignment lag of A); under a steady field at an interface it reshapes the *drop* (deformation of B); under an oscillating field the alignment lag of A re-enters and *governs* the drop's shape response (C). The through-line is the magnetization lag: a 54° delay measured three independent ways — in the bulk M–H loop (A), and inside the droplet both in the time trace and

in the frequency dependence (C) — all agreeing. The solver reproduces the validated bulk rheology (A vs Bacri) *and* gives physically sound, mass-conserving, convergent interface results (B, C).

Open directions. (i) Push B3 to higher M_s to locate the break-up threshold. (ii) Map the B2 crossover at several field strengths (does the balance F_b scale as predicted with Bo_m?). (iii) Couple the two sections: does the A-type alignment *lag inside the drop* modify its deformation dynamics under an oscillating field?

Caveat. B results are at modest resolution (R=10 on 50×50, interface $\sim$4 cells). The *qualitative* physics — monotonic field deformation, the competition minimum, the orientation flip — is robust; exact numbers and the precise crossover location would sharpen on a finer grid.